S09. Stoquasticity under a bounded class of basis changes

Pure Thought research portfolio

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Source: synthesis challenge 25; original synthesis. This topic is distinct from, or adds a distinct objective to, the canonical PRD catalog. Crosswalk

Plain-language guide

The problem in everyday terms

Quantum calculations can involve large positive and negative contributions that almost cancel, making some sampling methods inefficient. Sometimes a change in the mathematical basis removes troublesome signs. This problem asks whether that is possible within a finite, explicitly allowed set of local basis changes, so that both success and failure can be checked completely.

Key terms

- **Basis** — The reference states used to write a quantum operator as a matrix.
- **Hamiltonian** — The matrix describing the system's energy and couplings.
- **Off-diagonal entry** — A matrix element linking two different basis states.
- **Stoquastic** — Having real, nonpositive off-diagonal Hamiltonian entries in the specified basis.
- **Sign problem** — A sampling difficulty caused by signed or complex weights and cancellation in a chosen representation.
- **Pauli operator** — One of the standard single-qubit matrices used to express spin interactions.
- **Clifford transformation** — A basis transformation that maps Pauli operators to Pauli operators under conjugation.
- **Product transformation** — A change of basis applied independently to the individual qubits.
- **Satisfiability certificate** — A checkable assignment meeting encoded constraints, or an appropriate proof that none exists.

Why this matters

Knowing when signs can be removed helps separate difficulties caused by representation from those that survive a proposed remedy. A bounded search also creates small, exact benchmarks for algorithms intended to find useful basis changes.

What progress would mean

A successful result would supply and verify a transformation, or exclude every transformation in the allowed finite class. The complete Hamiltonian must be checked because contributions can cancel. Stoquasticity alone does not guarantee fast sampling, and failure for product Clifford changes does not exclude all possible bases.

Revised core question

For a finite rational Pauli Hamiltonian, is there an allowed product Clifford transformation making all off-diagonal entries real and nonpositive?

Precise scope, target and outputs

Fix n qubits, rational coefficients, and allowed U=⊗_iU_i with U_i from the finite single-qubit Clifford group modulo phase. Define stoquasticity in the computational basis for the full transformed H, not automatically term by term. Exhaustively enumerate or use an exactly equivalent constraint encoding. Export U and an exact matrix/sign check if successful; otherwise export complete enumeration or a checked UNSAT proof. Continuous local unitaries require a semialgebraic formulation and are a distinct extension.

Detailed critique of the synthesis framing

The original omits which local basis changes are allowed, so "unavoidable sign problem" has no definite quantifier. A SAT search over finitely many transformations cannot exclude all continuous local rotations. Requiring every term to be stoquastic can also be stronger than requiring their sum to be stoquastic because entries can cancel. Finally, stoquasticity does not guarantee efficient sampling or mixing. The rewrite makes a finite, falsifiable basis-cure problem and limits every no-go conclusion to its transformation class.

Validation and milestones

Baseline: exhaustive small-n decisions with explicit transformations or coverage. Strong: verified SAT encoding checked against brute force. Extend to a continuous rotation class only with exact equations, inequalities, and global proof machinery.

Use the shared evidence standard. Unsuccessful search is unresolved unless a complete exclusion proof is supplied. These are proposed baseline specifications, not completed scientific results.

Marvian, Lidar and Hen, On the computational complexity of curing non-stoquastic Hamiltonians supports the need to specify the allowed basis class.

Displayed formulation and claim boundary

Supplemental specification.

$$U = \bigotimes_{i=1}^{n} U_i, \quad U_i \in \mathrm{Clifford}_1/\mathrm{phase}, \qquad H' = UHU^\dagger,$$

$$\mathrm{Im}\, H'_{xy} = 0, \qquad \mathrm{Re}\, H'_{xy} \le 0 \quad (x \neq y).$$

Apply these conditions to the full transformed matrix, allowing cancellations between terms. Exhaustive coverage or an equivalent checked UNSAT proof excludes only this finite transformation class. Stoquasticity alone does not guarantee efficient mixing or sampling.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.